NCVA CRYPTOLOG, Corvallis, Oregon, Winter 1990
(Naval Cryptologic Veterans Association)

The Siemens Cipher Teletype in the History of Telecommunications

Today when one hears the word *America* in discussions of countries that have produced great inventions in communications technology what people associate with it are names like Samuel Finley Morse for the telegraph, Alexander Graham Bell for the telephone, and John William Mauchly for the computer. All this is a one-sided view, however, and does not do justice to any serious history of technology. It is true that the men I have just named have made widely known and immortal contributions, but in regard to the title of *the inventor* or *the first* we must differentiate.

It was from this point of view I welcomed and supported the article by Dr. Jarotskiy of the Post Office in Moscow in remembrance of the 150*th* birthday of Johann Philipp Reis, born 7 January 1834 in the Barbarossa town of Gelnhausen, as a *pioneer in telephony* in No. 1/1984 of the technical journal *Electrical Communications*. The author made use of a quotation, you see which I would like to share with you.

It runs: "A Critical history of technology would generally show that hardly any invention of the 18*th* and 19*th* centuries belongs to one individual or another."

Incidentally, this quotation comes from, a gentleman who is known to all of you, he is the jurist, philosopher and utopian Karl H. Marx.

I think we must agree with Marx, because his comment also holds true to a remarkable extent in regard to the development of telegraph equipment.

The processes of the development of telephone and computer technology also bring up a multitude of names, which I will not enumerate here.

Instead, let us stay in the realm of technology. Its rapid development was based on discoveries by scientists whose names, gentlemen, you know as well as I do. I remind you of Galvani, Oerstedt, Faraday, Henry, Ohm, and Ampere, to name only a few.

But how did it begin?

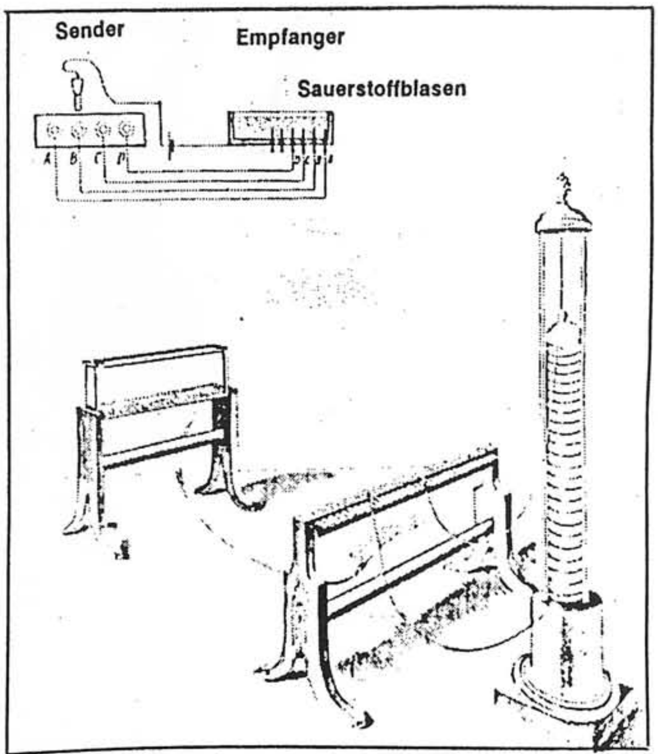
On the question of origin in the field of *pre-Philipp Reis* electrical telegraphy, I may emphasize its germ in Germany, namely the *electrochemical telegraphy* of the physician and member of the Royal Academy of Sciences at Munich, Samuel Thomas von Soemmerring, born in 1755 at Thorn.

After the invention of the electrical pile by the Italian physicist Volta in 1800, Soemmerring succeeded in setting up an impressive but slow parallel-wire telegraphic system over a distance of around 1000 meters, making use of the releasing of gas. Soemmerring demonstrated his method in Munich at the Academy in 1809, but he was unable to make any headway against the *optical telegraphy* (i.e. semaphore) which was dominant at that time, (e.g., the Chappe System, used by Napoleon Bonaparte at that time). However, his labors were of inspirational value for the later telegraph systems that Schilling von Canstatt and Moritz Jacobi developed in Petersburg. They determined the further progress of two-wire telegraphy. For *electrical telegraphy*, in Russia 21 October 1982 was the 150*th* anniversary of the invention of the electromagnetic telegraph, the day when the German-Russian Baron Paul Schilling von Canstátt successfully demonstrated his six-pin apparatus for the Russian alphabet to the Czar and the Academy in Petersburg in 1832. After his death in 1837 his work was continued until 1874 by Jacobi, a Prussian master builder in Koenigsberg and from 1837 on a scientist in St. Petersburg.

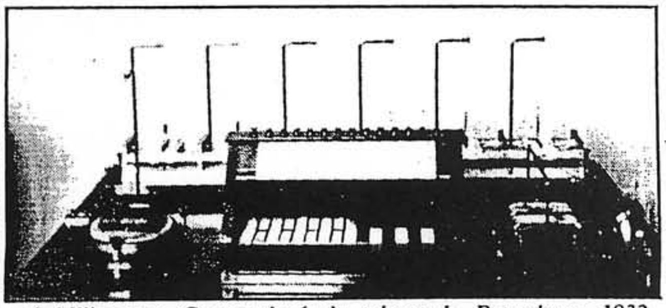
Let us not trouble now to go into the numerous events in the evolution of telegraphy from 1833 to the start-stop teleprinter of the 1920s in detail. But emphasis should be given to the essentials. Some systems established milestones and other sooner or later were replaced. Thus, for example, the needle telegraphs did not achieve lasting prevalence, whereas the American painter Samuel Morse, in the years from 1837 to 1844, in creating so-called recording telegraphs and inventing the alphabet that is named after him made an inestimable contribution that is still influential at the present time. Morse gave credit publicly to the German Steinheil, who first proved to use the ground as the return loop, which saved the installation of a *second* wire and eased the introduction of the Morse system in the US Continent in 1844.

Still more successful was the technology of the type printing telegraphs that came into use in 1854. A well known representative of this type of equipment is the so-called Hughes printer, a very remarkable device with a life in service of almost 8 decades.

I have been told that there are several members represented in your Association who were trained on this machine here in Muerwik at the Torpedo Communications School before the beginning of the 1930s. If you should ever come to Frankfurt am Main some time, I recommend to you a visit to the Federal Postal Museum. There on display are two different models of the Hughes printer, in working condition. In 1900 there were already 3500 of his teletypes in existence in the whole world when Edward Hughes dies a millionaire and one of the leading consultants in telegraphy. As early as 1895, Hughes printers were being developed further at Siemens, and by the beginning of the teletype generation of the 1920s they were being built. A parallel development in 1903, that of the Siemens automatic telegraphs with a speed of 2000 characters per minutes, (i.e., 33 characters per second), was far in advance of its time. It was followed in 1924 at Siemens by the 56 Baud keyboard printing telegraph, which still came in two pieces (i.e.,



Soemmerring's electrochemical telegraph. Munich, 1809. Later improved by Schweigger for use over only two wires, 1811 (but not realized).



Schilling von Canstatt's 6-pin telegraph. Petersburg, 1832. Courtesy: Deutsches Museum, Munich.

separate sender and receiver. See picture). That became the starting point for the electrical teletype. You see, there was more demand for a machine that would send, receive, be operated by only one person, and still be cheap as well. What was aimed at was a machine similar to the type bar carriage typewriter in common use at that time, which had been factory made since the second half of the 19*th* century. It was in this way that the start-stop teleprinter patents of the German-American Krum originated in the USA, financed by Morton and with the trademark MORKRUM, and in 1923 for Kleinschmidt, a German-American from Bremen. The two firms merged in 1924 to form the Morkrum-Kleinschmidt Corporation. Next came the development of the "Teletypewriter Model 14," which was imported under the name *Springschreiber* (i.e., start-stop teleprinter) in 1926 by Lorenz of Berlin. Lorenz copied the "Teletype 14," and Siemens & Halske successfully met the challenge from America too. Between 1927 and 1933 Wernerwerk of Berlin likewise developed a series of different teletype machines in rapid succession. In the course of doing so, a transition was made from steps under patent to taking new directions.

In 1927, when the German licensee of the principle of the *lengthened stop signal* in the mechanical teletype, patented by Kleinschmidt, would not share his monopoly with other manufacturers, F. Geissler, a designer in our firm, had the idea of cleverly circumventing the Kleinschmidt patent through an *electrical* plan. There came into existence an electrical teletype machine, "T type 25", whose further development -- available in 1932 -- is familiar to all of you gentlemen. I refer to the T of "T type 36." Until now it remained unclear why it was this particular machine, which later served as the basic module for a *secret teletype machine*, that attracted the interest of the Navy in the first place. The Navy preferred the electrical machine over the mechanical machine because of its superior stability at sea.

This machine superseded the Hughes printer that had been in use up to then, and the *sounder*. The change from the piano-like keyboard to a completely new set of keys may have caused a few problems at the time. Perhaps one or another of you can confirm that.

Of the further developments that followed at Siemens, I would like to point out the following mass-production teletypes, all of them page printers:

- 1933: T type 37a (successor to the T type 26a of 1929),
- 1957: T 100, 50 Baud capacity
- 1961: T 100, 100 Baud capacity,
- 1976: T 1000, already highly integrated.

The following figures point to the worldwide significance of these developments:

By 1984 Siemens had supplied over 1 million teletype machines and had become the unquestioned market leader among 13 manufacturers.

Our firm's share in the total number of 1.6 million telex machines in the world is 30%.

By the end of the second World War, 15,000 machines had been produced; in recent years, 1978 to 1983, it was 70,000 machines a year, just on the average.

From this point of view of types of technology, we get the following distribution figures:

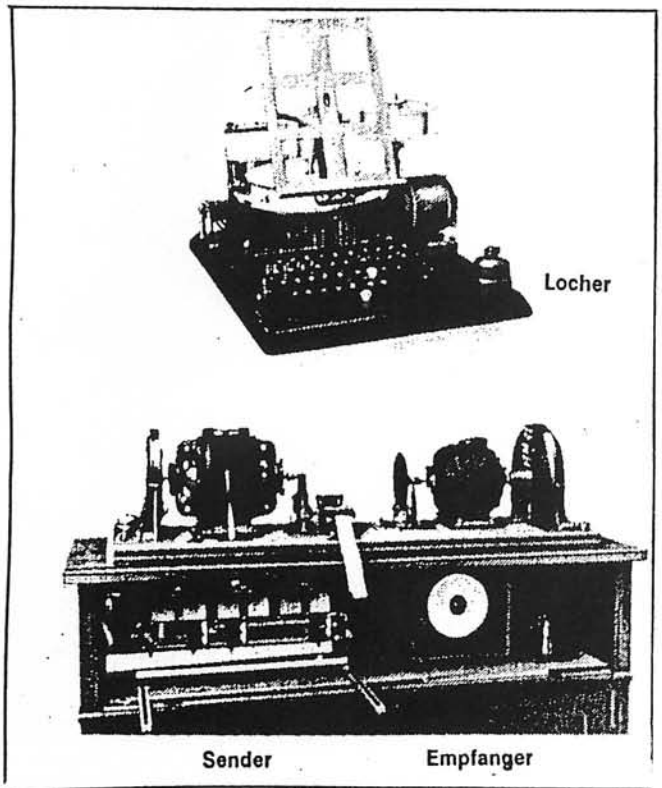
- 55% electromechanical, including Model T100
- 40% electronic, first generation
- 5% electronic, second generation, microprocessor controlled.

After this short outline of the history of the development of telegraphy, I would like to take up a machine that you all know, gentlemen, and which even yet -- almost 40 years after the end of the war -- is surrounded by secrets. I mean the Siemens cipher printer "T type 52", models "a" through "f". On the construction and manner of operation of this machine I do not need to say much. You, gentlemen, have worked with this machine during your time of active service. Many of you have been at Siemens in Berlin, and have taken part in the course of instruction there. Instead of going into that, allow me to make a few remarks on the development of this machine and the enemy efforts at decipherment.

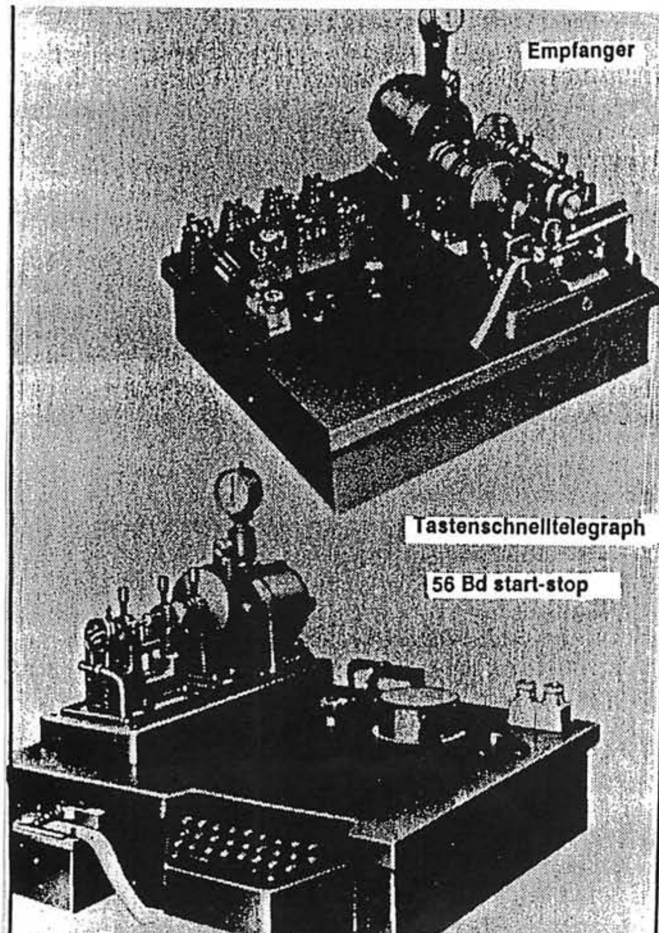
In 1969 a colleague in our firm, Ehrhard Rossberg, director of the Technical Division of Switching Technology of Siemens retired. The patent department marked the departure of this extraordinarily skillful engineer with a separate volume which contained more than 100 of his patent descriptions. Mr. Rossberg, who was born in 1904, attended the Chemnitz Industrial Academy after a period of *practical training, and came to Siemens & Halske in Berlin-Siemensstadt in 1926.*

At Siemens, Mr. Rossberg was on the scene at the new starting point of the T type 25 electrical teletype and in 1928, when the Navy in Kiel selected it. A productive collaboration ensued shortly with additional, very successful colleagues such as Honorary Dr. of Engineering Herbert Wuesteney, Honorary Dr. of Engineering August Jipp, Eberhard Hettler, Fritz Geissler, and others.

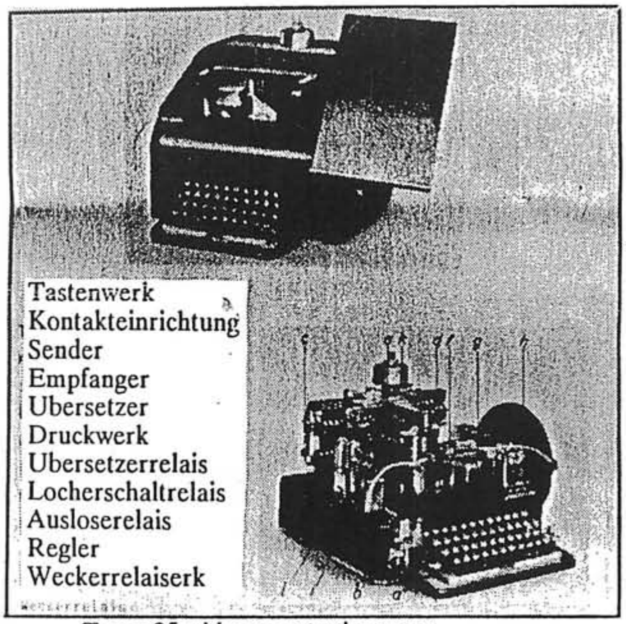
The Navy, concerned because of the British decipherment successes in the first World War, made it a requirement that the electrical teletype should be provided with an enciphering device.



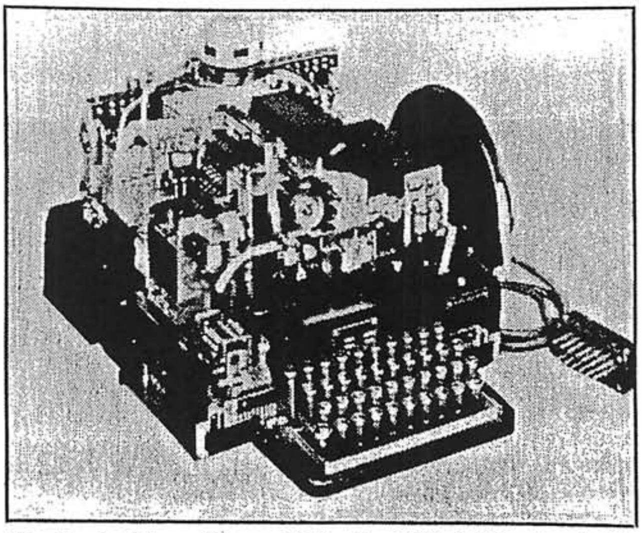
Siemens high-speed telegraph. Berlin, 1903. Courtesy: Siemens Museum, Munich,



Siemens keyboard printing telegraph. Berlin, 1924.

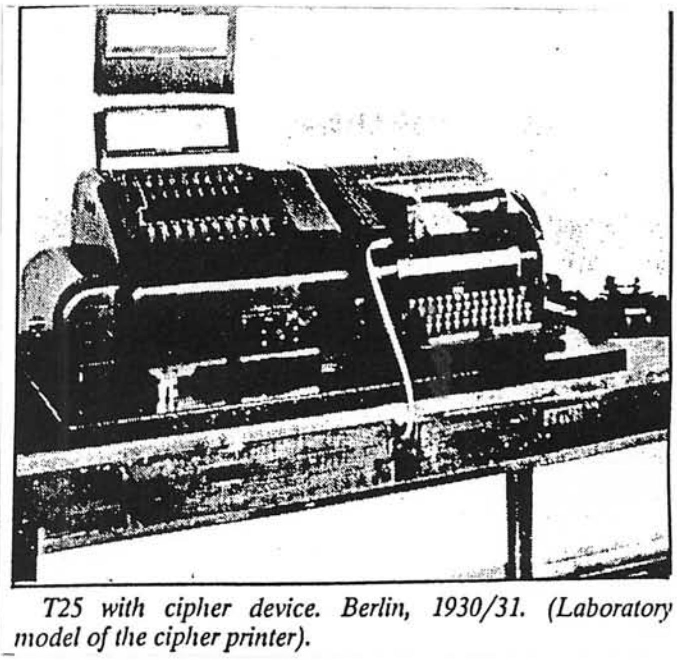


T type 25 without protective cover.



Electrical teletype, T type 36. Berlin, 1932 (without casing).

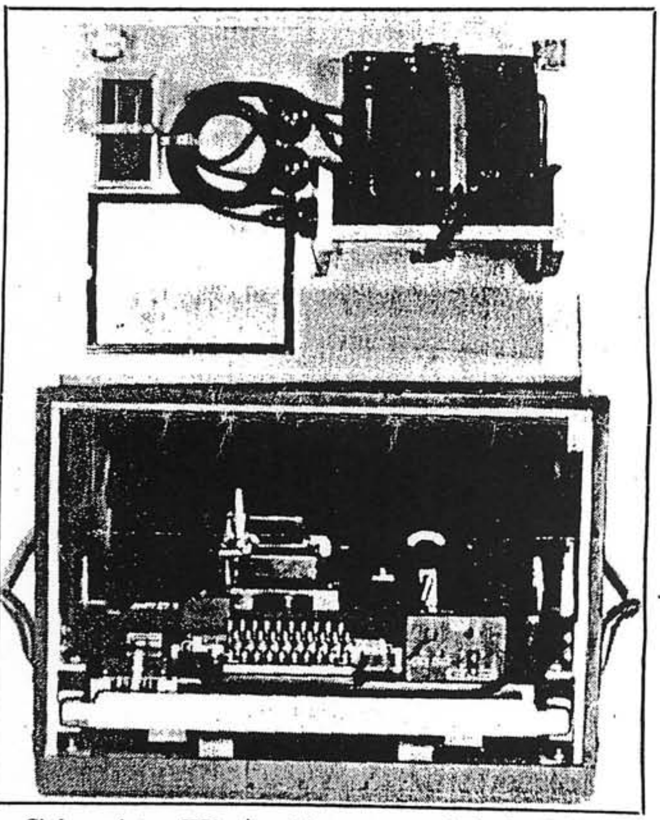
This idea was unusual, because at that time teletype text requiring protection were enciphered before transmission in a fussy and time-consuming manner. But what they had in mind now was the truly sensational development of what today is called an on-line machine. It is possible also to ascribe this to the design advantages of a the electrical teletype, whose creation was provoked (i.e., by the Kleinschmidt patent problem), but was fortunate.



T25 with cipher device. Berlin, 1930/31. (Laboratory model of the cipher printer).

On the way there, we come in 1930 to the first applications for patent with the title *Arrangement of Switching for the Transmission of Information in Cipher*.

A further application from the year 1931 contains the principle of irregular cam discs, i.e., cam discs of different sizes, set up chiefly on the basis of prime numbers. In the same year the patent application was filed in the USA by Jipp, Rossberg, and Hettler, there US patent 1,912,983 was granted 6 June 1933.



Cipher printer, T52a/b, with power supply in the lid, before WWII.

It has been reported by severa;l veterans that the Navy was the first service branch to be interested in using the cipher teletype machine, also called a *cipher printer (Geheimschreiber)*, of *G-Schreiber* for short. The Navy wanted secure written communications, for example, from a ship at the pier to the high command of the fleet.

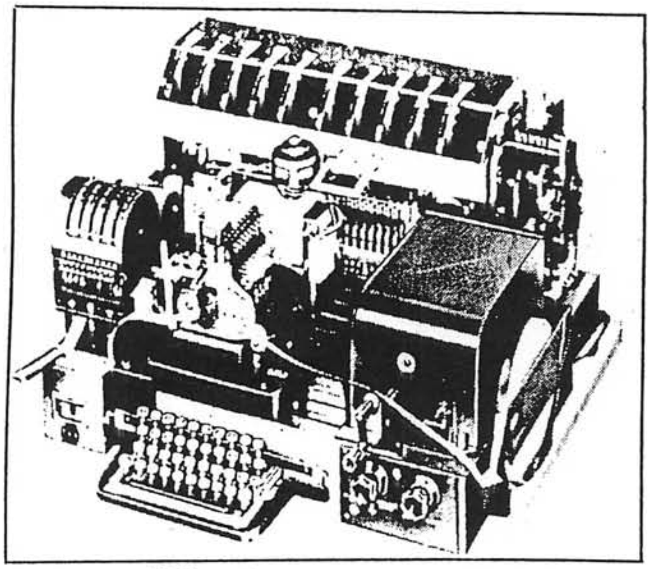
I owe it to a fortunate circumstance that in 1982 in Munich I was able to discover -- or rather rescue -- a photograph that shows the laboratory model from this first couple of developmental years. It still consisted of two separate machines, the T type 25 electrical teletype and the cipher device with 2 x 10 adjustable wheels. This *attachment* was later, in the commerical version of 1932, united with the T36, in *one unit*, the "T52".

Now we go to the introduction of the Siemens T type 52a cipher teleprinter.

In 1932 the Hungarian armed forces were the next to be supplied with machines and complete documentation.

Navy tests soon showed that the cipher printer interfered with the sensitive radio receivers. For this reason the model T type 52a/b with noise suppression circuitry was created in 1934.

A further step in the development of the cipher printer was the T type 52c, for which the Air Force gave specifications in 1936.



Cipher printer, T52c, without casing.

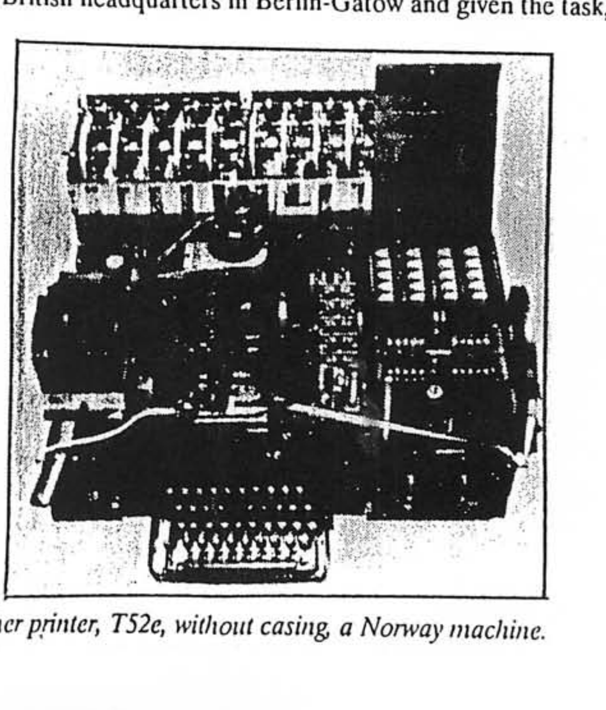
In the meantime, the security authorities of the Third Reich imposed secrecy measures which still make themselves felt today in the considerable scarcity of information.

Nevertheless, after the signing of the German-Russian treaty on 28 September 1939 in Moscow, the Siemens SFM T52 cipher teletype machine was demonstrated and shown opened up to Soviet military delegation in Berlin-Siemensstadt. (By order of the Reichs Government).

Development continued during the second World War. Models T type 52d and T type 52e were created, and towards the end of the war Model T type 52f was in the works, at least in specification form.

In 1944 or '45 the technical documentation in the factory for the strictly secret T52 became lost. For one thing, the production facility in Berlin was badly hit by Allied bombs in October 1944. But what stands today as the main reason for unsuccessful endeavors to obtain authentic records and technical descriptions in the rigorous security regulations and their application before the occupation in 1945.

After the end of the war Dr. Wuesteney was brought to the British headquarters in Berlin-Gatow and given the task,



Cipher printer, T52e, without casing, a Norway machine.

together with Fricke, a Siemens mechanic who worked on T52 assembly, of reconstructing the circuit diagram of a T52. At the end of a week the British discontinued the project without result, although they are said to have gotten possession of no documents, but only machines, which Mr. Fricke, among others, serviced. The machines were later put to use on teletype connections between the British Sector in Berlin and the headquarters of the Army of the Rhine.

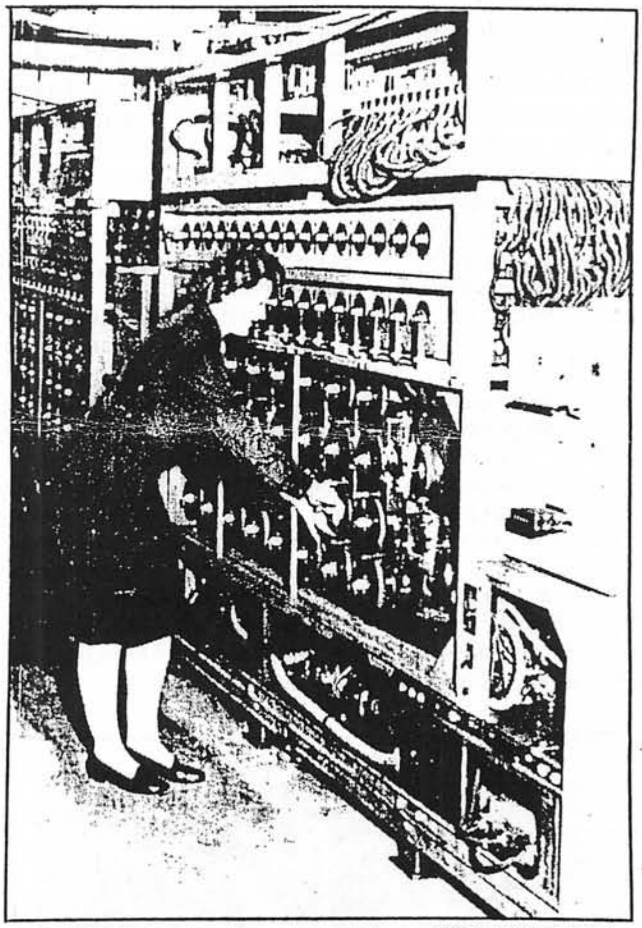
In Norway, too, a number of T52c's survived the end of the war, and were later put into use again there. Of these there are speciments in operating condition on display today in the Siemens Museum in Munich, in the Science Museum in London, and in the Technology Museum in Oslo. The exhibit in the Military Museum at Dieppe-Pourville on the other hand, a "d" (in phonetic alphabet Dora) machine (T52d), which I saw last month, is incomplete, poorly maintained, and hence not in working condition, although the T rls 43a telegraph relays, for example, look like new.

From the former armed forces' telecommunications equipment depot which was in Elmshorn between 1948 and 1960, the remaining cipher printers, which had previously been demilitarized, i.e., partly dismantled, were brought to Trier and there at the *Electromechanical and Electronic Plant* were put in circulation, as the French security authorities had decided on utilization of the cipher printers. Trier supplied a total of 280 T52d's and e's, which is an estimated 25% of all T52s supposed to have been produced in Berlin up to 1944. The cipher printer was classified down to the present time, and may still be in use somewhere. Investigations become all the more difficult; informants prove to be a decided rarity.

This may be sufficient at this point as a presentation of the development of the cipher printer.

But now to the question of the extent to which the enemy was able to break into our machine's cipher texts. I can anticipate the answer: Up to the present time we have not been able to obtain any reliable information -- but let us neverthless attempt an approximation.

Soon after the first opening of the British secret records in 1975, 30 years after the second World War, the British journalist Brian Johnson wrote in his book *The Secret War* about technical facilities of the British decipherment center at Bletchley Park. According to that book, there were the so-called bombes for processing the Enigma material, and forerunners of the computer, designated *Colossus,* were available for deciphering the cipher printer texts. This statement went from one book to another, and was even broadcast by the BBC in both words and pictures.



Photos courtesy NSA

Bombe equipment installed at 3801 Nebraska Avenue, Washington, D.C.

Knowledge of our own and indications from England over the past years have confirmed us more and more in the belief that the above-named author's investigations were in behalf of the Siemens & Halske Geheimschreiber inadequate. Recently our belief has become a certainty. Because the third volume of the official British history of the operations of the intelligence service in the second World War has been published. In it, in Part I, of Volume 3 the principal mission of the *Colossus* decipherment computer is described: Actually, it was used in decrypting German 5-character code messages. These messages however came not from the Siemens cipher printer, but rather from another teletype cipher printer, the model SZ40 and SZ42 cipher attachment from the firm of Lorenz -- a machine, therefore, whith which you may have had less contact, gentlemen.

In order to substantiate this statement further, at this point let a short review of the development of radio teletype technology be permitted.

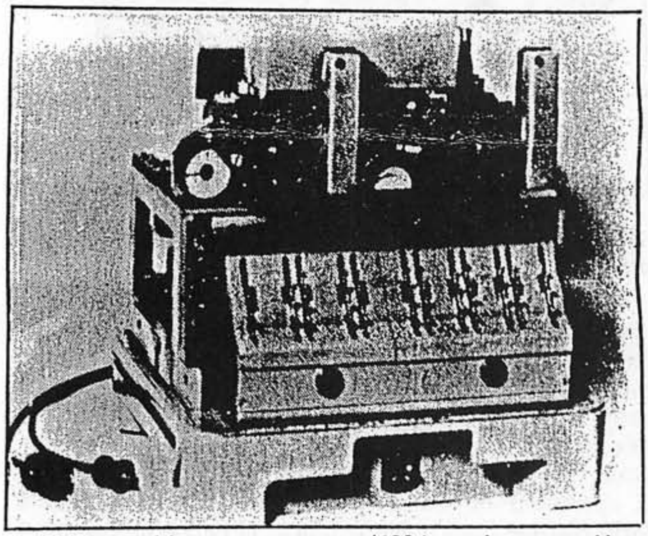
As a result of the constantly increasing distances between the fronts, in 1940 a need grew up in the German armed forces for a serviceable radio teletype system that cipher equipment could be incorporated into.

To meet this need, in the course of four years a series of various devices was developed and in varying quantitites was requested and operated by users, chiefly the Army. The first of these shortwave radio teletype sets with telegraphy channels went by the covername of *Saegefisch* (i.e. Sawfish) on the German side, by the way. The British intelligence service found out this designation from a decrypted Enigma message, and for their part used the codeword *fish* for it. Since about 1978, this covername has been incorrectly applied to the Siemens cipher printer, and indeed even when in reality what is being discussed is the German 5-character code traffic in general, or even the decipherment of Army communications done on the Lorenz cipher attachment.

It would be dishonest to the history of the field if it went unmentioned that the T52 designed by Siemens for wire communications lines was also used in its "b" and "c" models for the first radio teletype trials ([19]40/41). It soon turned out, however, that both models were suited only for limited use. This was because if it happened, for example, that as a result of atmospheric disturbances even a single letter of cipher text was suppressed, then the encipherment's synchronization was thrown off and subsequent text was decrypted incorrectly. The problem was overcome, however, with the Lorenz machines already mentioned.

Thus when the British systematically monitored the German experimental radio teletype link between Vienna and Athens in 1941, they thought they were on the trail of the Siemens cipher printer. We must even concede today that British mathematicians had then already succeeded, by awkward manual procedures, in deciphering German teletype messages. In so doing, they realized that only automatic decrypting machines could give effective support to their own military operations. These were actually put into use in June 1942, and had to be constantly modified because of their severe rate of wear and breakdown.

The British were more successful with the Colossus computer, which was not put into use until February 1944, however. An example on this point:



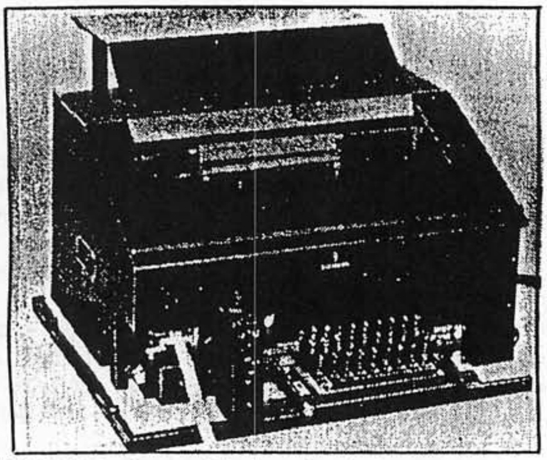
SZ42, without cover. (1984 pictures, Navy Telecommunications School, Flensburg).

Beginning on 6 January 1944 the radio teletype link belonging to the Commander in Chief in the west (Army Group D) between Paris and the OKW [Supreme Command of the Armed Forces] in Berlin Straussberg radio station) was intercepted, and starting in March, about three months beore the beginning of the invasion, they were in a position to decrypt teletype enciphered on the SZ42 *(Schlvesselzosatz 42)*.

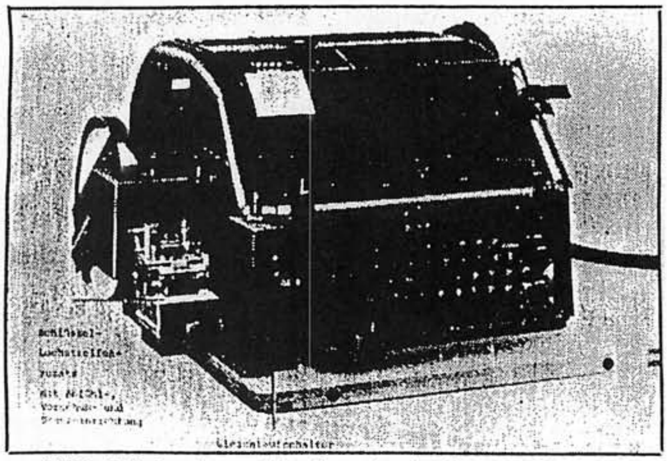
But the "d" model of the Siemens cipher printer was also intended for radio teletype operation. Here is the proof: It is from among you, gentlemen, that valuable training materials from the Naval Communications School at Waren-Mueritz dating from 1944 have been made available to the Archive of the Naval Telecommunication School. They contain operating directions for the machine just mentioned. So far, however, it has remained unknown whether the Navy has at any time ever operated communications as we see them shown here. It would be very nice if this matter could be clarified with your kind assistance.

With that I come to my conclusion:

I believe it has become clear that we are only at the beginning of the truth. The official British historiography has up to now not been prepared to make any statement regarding the extent of the decipherment of teletype texts. Might this perhaps be because they would have to admit to not having had unqualifiedly splendid succcess? I ask you to consider: Up to the present time we still have no information on how the attempts at decipherment turned out in so far as they relate to the T type 52 "d" and "e" cipher printer of the "SFM T43" cipher teletype machine also developed by Siemens late in World War II. Wernerwerk T in Berlin built the "T43" on the basis of two types of teletype machine. These were the T type 37f mechanical page printer and the T type 34n mechanical tape printer, of which a hand full of machines started working in 1944 on a one-time-pad principle, e.g., between Paris and Berlin.



SFM T43, basic model of mechanical page printer, T type 37f.



SFM T43, basic model of mechanical tape printer, T type 34n.

Thus for the present the question remains open as to the extent to which the teletype decipherment added to the significance of the Enigma decipherment or qualified its value. In the same way it must be kept for a later historical study to elucidate the degree to which the decipherment successes that have been presented influenced the strategy and tactics of the Allied invasion.

WORDS USED

1.		*Sender*	transmitter
		Empfaenger	receiver
		Sauerstoffblasen	oxygen bubbles
3.		*Locher*	perforator
4.		*56Bd*	56 Bauds
5.	a.	*Tastenwerk*	keyboard unit
	b.	*Kontakteinrichtung*	contact device
	c.	*Sender*	transmitter
	d.	*Empfänger*	receiver
	e.	*Übersetzer*	converter
	f.	*Druckwerk*	printing mechanism
	g.	*Übersetzerrelais*	converter relay
	h.	*Locherschaltrelais*	perforator switching relay
	i.	*Auslöserelais*	release relay
	k.	*Regler*	control
	l.	*Weckerrelais*	relay controlling local bell circuit
12.		*Gleichlaufeinrichtung*	synchronizing device
		Funkfernschreibbetrieb	radio teletype operation
14.		*Abfühleinrichtung*	tape reading device
		Vorschubeinrichtung	feed device
		Stanzeinrichtung	punching device
		Gleichlaufschalter	synchronization switch

BIOGRAPHICAL SKETCH

Wolfgang W. Mache, member of VDE/ITG, derived his interest in WWII cryptology from Brian Johnson's book *The Secret War* (1978), whilst he wrote the *Lexikon der Text- und Datenkommunikation* (1980 Oldenbourg). He joined Siemens for telephone technician training, received a Dipl. Ing. degree in Electricial Engineering (1955) and then worked in Siemens' Telecom development Centers, Munich-Sendling for Power-line telephony and FDM/TDM-ARQ radio teleprinter systems, a time of six years as Telecom Sales Manager in South America, Lima/Peru, he is in marketing in the field of data transmission and switching.

Photo by Hannelove
Wolfgang W. Mache

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By Wolfgang W. Mache

CRYPTOLOG Editor's note: This article was originally printed in CRYPTOLOGIA, and is reprinted with their permission, and the permission of the author. The author made several corrections to the original. Wolfgang Mache has also submitted an independent article to CRYPTOLOG which will be printed in a future issue.

This is a translation of a speech given by Mache at the 23rd Federal Meeting (Bundestreffen) of the KAMERADSCHAFT DER MARINEFERNSCHREIBER under the auspices of the MARINEFERMELDESCHULE FLENSBURG-MVERWIK, which is next to the Flensburg Naval Base. It was the third meeting of the German Naval Teleprinter Veterans Association, which was held on the 700th anniversary of the town of Flensburg. A number of former MARINE-NACHRICHTENHELFERINNEN of World War II also attended.

The Siemens Company, Munich, was represented by engineers Karl-Heinz Schramm and the author Wolfgang Mache.

These Siemens engineers were aware that the German Naval Service was the first and most important customer of the first Siemens Electrical Teleprinter developed in 1928, and of the Siemens-Geheimschreiber T52a, before the Third Reich appearred. A T52e in working order was donated to the schools SAMMLUNG, or collection of equipment dating back to the Reichsmarine. The meeting was held under the auspices of K.z.S. Klaus Ehlert. Technical assistance was given by Kapitaenleutnant Joachim Timm. Timm retired in 1987 from the Bundesmarine. Kapitaenleutnant Timm is currently researching the probability that the German Navy used a typewriter-type cipher unit prior to 1914.*

*ANT hlm

Control Number 1135

STAFF INFORMATION SECRET!

The Siemens SFM T52d Cipher Teletype Machine

SECRET STAFF INFORMATION!

1. This is a state secret within the meaning of Section 88 of the Reichs Criminal Code.

2. To be forwarded only by hand carrying or addressed by name, in double envelope, and upon written acknowledgement of receipt.

3. Transmission by courier or by a trust person if possible; with a declared valuation of more than 1000 Reichsmarks if mailed.

4. Reproduction of any kind or taking extracts is forbidden.

5. Storage is upon the responsibility of the recipient, to be in a safe or under exceptional conditions in a steel locker with a combination lock.

6. Violations of these rules will be most severely punished.

Printed Naval Regulation No. 35 IV

Cover sheet for the Siemens SFM T52d Cipher Teletype Machine.